INFOSYS SE ASSESSMENT 2026 • OFFICIAL PREPARATION VAULT

🔢 Section 2: Mathematical Ability — Timed Sectional Mock Tests (10 Questions | 35 Minutes)

Section: 09_Mock_Test_Series | **Drive Date:** 5th & 6th November 2026 | **Status:** 100% Solved & Verified

🔢 Section 2: Mathematical Ability — Timed Sectional Mock Tests (10 Questions | 35 Minutes)

Official Pattern: Section 2 of Infosys Systems Engineer (SE) Exam
Allotted Time: 35 Minutes (Countdown Timer)
Question Count: 10 Questions per Mock Paper
Passing Benchmark: 7+ Marks (70% Accuracy)
Negative Marking: Zero (0)

📝 SECTIONAL MOCK TEST 01 (10 Questions | 35 Mins)

Question 1 (Time and Work - Alternate Days)

A can complete a piece of work in 12 days and B in 18 days. If they work on alternate days beginning with A, in how many days will the work be completed?

- A) 14 days
- B) 14.33 days
- C) 14.5 days
- D) 15 days

Correct Answer: B

Solution: Total work = $\text{LCM}(12, 18) = 36$ units. Efficiency of A = 3 units/day, Efficiency of B = 2 units/day. In a 2-day cycle, work done = $3 + 2 = 5$ units. After 7 cycles (14 days), work done = $7 \times 5 = 35$ units. Remaining work = $36 - 35 = 1$ unit. On day 15, A works: time taken = $1 / 3$ day. Total time = $14 + 1 / 3 = 14.33$ days.

Question 2 (Profit, Loss and Discount)

A shopkeeper marks his goods 40% above the cost price and allows a discount of 25% on the marked price. In addition, he gives 1 article free with every dozen purchased. What is his net profit or loss percentage?

- A) 2.5% loss
- B) 3.07% loss
- C) 5% profit
- D) No profit no loss

Correct Answer: B

Solution: Let CP of 1 article = ₹100. CP of 13 articles = ₹1300. MP of 1 article = ₹140. For 1 dozen (12 articles),

customer pays for 12 articles at 25% discount: $SP = 12 \times 140 \times 0.75 = 12 \times 105 = ₹1260$. Net profit/loss = $1260 - 1300 = -40$ (Loss). Loss % = $40 / 1300 \times 100 = 3.07\%$.

Question 3 (Permutations and Combinations)

In how many different ways can the letters of the word `CORPORATION` be arranged so that all the vowels always come together?

- A) 50,400
- B) 28,800
- C) 14,400
- D) 7,200

Correct Answer: A

Solution: Word `CORPORATION` has 11 letters: C, R(2), P, T, N (6 consonants) and O(3), A, I (5 vowels). Treat 5 vowels as 1 single block. Total items to arrange = 6 consonants + 1 block = 7 items with R repeated twice: $7! / 2! = 5040 / 2 = 2520$. Within the vowel block, 5 vowels with O repeated 3 times: $5! / 3! = 120 / 6 = 20$. Total arrangements = $2520 \times 20 = 50,400$.

Question 4 (Time, Speed and Distance - Relative Speed)

Two trains of lengths 140 m and 160 m run on parallel tracks. When running in the same direction, the faster train passes the slower train in 30 seconds. When running in opposite directions, they pass each other in 6 seconds. What is the speed of the faster train in km/h?

- A) 90 km/h
- B) 108 km/h
- C) 126 km/h
- D) 144 km/h

Correct Answer: B

Solution: Total distance = $140 + 160 = 300$ m. Same direction: $u - v = 300 / 30 = 10$ m/s. Opposite direction: $u + v = 300 / 6 = 50$ m/s. Adding equations: $2u = 60 \Rightarrow u = 30$ m/s. Speed in km/h = $30 \times 18 / 5 = 108$ km/h.

Question 5 (Compound Interest)

The difference between the compound interest and simple interest on a certain sum of money invested for 2 years at 8% per annum is ₹192. What is the principal sum?

- A) ₹25,000
- B) ₹30,000
- C) ₹32,000
- D) ₹35,000

Correct Answer: B

Solution: For 2 years, $\text{Difference} = P\,(R / 100)^2$.

$192 = P\,(8 / 100)^2 = P\,(64 / 10000) \Rightarrow P = (192 \times 10000) / 64 = 3 \times 10000 = ₹30,000$.

Question 6 (Pipes and Cisterns)

Two pipes A and B can fill a cistern in 20 minutes and 30 minutes respectively. A third pipe C can empty the full cistern in 15 minutes. If all three pipes are opened simultaneously, in how many minutes will the cistern be

completely filled?
- A) 45 minutes
- B) 60 minutes
- C) 75 minutes
- D) It will never fill
Correct Answer: B
Solution: Capacity = $LCM(20, 30, 15) = 60$ units. Efficiency: A = $+3$, B = $+2$, C = -4. Combined net efficiency = $3 + 2 - 4 = +1$ unit/min. Time to fill 60 units = $60 / 1 = 60$ minutes.

Question 7 (Probability - Cards and Balls)

A box contains 5 red balls, 4 green balls, and 3 blue balls. Three balls are drawn at random without replacement. What is the probability that all three balls are of different colors?
- A) $3 / 11$
- B) $3 / 22$
- C) $5 / 22$
- D) $6 / 11$
Correct Answer: A
Solution: Total balls = $5 + 4 + 3 = 12$. Total ways to choose 3 balls = $\binom{12}{3} = (12 \times 11 \times 10) / 6 = 220$. Favorable ways (1 Red, 1 Green, 1 Blue) = $\binom{5}{1} \times \binom{4}{1} \times \binom{3}{1} = 5 \times 4 \times 3 = 60$. Probability = $60 / 220 = 3 / 11$.

Question 8 (Ratio and Alligation)

A merchant has 100 kg of sugar, part of which he sells at 7% profit and the rest at 17% profit. He gains 10% on the whole. How much sugar did he sell at 17% profit?
- A) 30 kg
- B) 40 kg
- C) 60 kg
- D) 70 kg
Correct Answer: A
Solution: By alligation: Ratio of (7% profit quantity) to (17% profit quantity) = $(17 - 10) : (10 - 7) = 7 : 3$. Total parts = $7 + 3 = 10$. Quantity sold at 17% profit = $3 / 10 \times 100 = 30$ kg.

Question 9 (Number Systems - Divisibility & Remainders)

Find the unit digit of the expression $7^{95} - 3^{58}$.
- A) 0
- B) 4
- C) 6
- D) 7
Correct Answer: B
Solution: Cyclicity of 7 is 4 ($7, 9, 3, 1$). $95 \pm od\ 4 = 3 \Longrightarrow 7^3 = 343$, unit digit is 3. Cyclicity of 3 is 4 ($3, 9, 7, 1$). $58 \pm od\ 4 = 2 \Longrightarrow 3^2 = 9$, unit digit is 9. Unit digit of $(13 - 9) = 4$.

Question 10 (Data Interpretation - Table Analysis)

A table shows the production (in thousands) of 4 types of cars by a company over 5 years:

Year	Sedan	Hatchback	SUV	EV
2021	45	60	30	15
2022	50	65	40	25
2023	55	70	50	45
2024	60	75	65	70

What was the percentage increase in EV production from 2021 to 2024?

- A) 266.67%
- B) 366.67%
- C) 400.00%
- D) 466.67%

Correct Answer: B

Solution: In 2021, EV = 15k. In 2024, EV = 70k. Absolute increase = $70 - 15 = 55$ k. Percentage increase = $55 / 15 \times 100 = 366.67\%$.